Introduction to Limits

Section 1: Intuitive Concept and Numerical Estimates

1 What does $\lim_{x \to c} f(x) = L$ mean intuitively?

a. As x gets arbitrarily close to c (from either side, $x \neq c$), $f(x)$ gets arbitrarily close to L

b. $f(c) = L$ always

c. The slope of f at c is L

d. $f(x)$ is continuous at c

2 Use numerical estimation with $x = 1.9, 1.99, 2.01, 2.1$ to estimate $\lim_{x \to 2} (3x - 1)$.

a. 5

b. 6

c. 4

d. 3

3 If $f(3)$ is undefined, can $\lim_{x \to 3} f(x)$ still exist?

a. Yes, because limits describe behavior near $x = 3$, not at $x = 3$

b. No, limits require $f(c)$ to be defined

c. Only if $f(x) = 0$

d. Only for polynomials

Section 2: Graphical Limits and One-Sided Limits

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4 When does the two-sided limit $\lim_{x \to c} f(x)$ exist?

a. If and only if both one-sided limits exist and are equal ($\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = L$)

b. Whenever $f(c)$ exists

c. Whenever $\lim_{x \to c^+} f(x)$ exists

d. If f is defined on $[c, \infty)$

5 For the step function $f(x)$ defined as 2 for $x < 1$ and 5 for $x \geq 1$, find $\lim_{x \to 1^-} f(x)$ and $\lim_{x \to 1^+} f(x)$.

a. Left limit $= 2$, Right limit $= 5$

b. Left limit $= 5$, Right limit $= 2$

c. Both equal 2

d. Both equal 5

6 Does the two-sided limit $\lim_{x \to 1} f(x)$ exist for the step function in the previous question?

a. No, because the left limit (2) does not equal the right limit (5)

b. Yes, it equals 5

c. Yes, it equals 3.5

d. Yes, it equals 2

Section 3: Algebraic Evaluation of Limits

7 Evaluate $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$.

a. 8

b. 0

c. 4

d. Does not exist

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8 Evaluate $\lim_{x \to 0} \frac{\sqrt{x+9}-3}{x}$.

a. 1/6

b. 1/3

c. 0

d. 1/9

9 Evaluate $\lim_{x \to -2} \frac{x^3+8}{x+2}$.

a. 12

b. 0

c. 4

d. 8

Section 4: Limits Involving Infinity

10 Evaluate $\lim_{x \to \infty} \frac{5x^2-2x}{2x^2+7x-1}$.

a. 5/2

b. ∞

c. 0

d. -2

11 Evaluate $\lim_{x \to \infty} \frac{3x+1}{x^2+5}$.

a. 0

b. 3

c. ∞

d. 1/5

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12 Evaluate $\lim\limits_{x \to 3^+} \frac{1}{x-3}$ and $\lim\limits_{x \to 3^-} \frac{1}{x-3}$.

a. $+\infty$ and $-\infty$

b. $+\infty$ and $+\infty$

c. $-\infty$ and $+\infty$

d. 0 and 0

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